Question 56

01: Use Copilot Responsibly

Two developers enter the exact same prompt into GitHub Copilot Chat in identical repositories, yet receive slightly different code implementations. What explains this behavior?

A. One developer has a corrupted IDE cache.

✔ **B. Large Language Models are inherently non-deterministic and sample tokens probabilistically based on context and sampling parameters.**

C. Copilot intentionally randomizes syntax to prevent code plagiarism.

D. Enterprise administrators configured A/B testing on user seats.

Correct Answer: B

Generative AI models predict token probability distributions rather than executing deterministic rule lookups. Minor variations in surrounding workspace state, open tabs, or probabilistic sampling can yield different—yet equally valid—code implementations.

Option Breakdown

- **A. Corrupted IDE cache:** Incorrect. Differing outputs are normal behavior for generative LLMs, not an indicator of software corruption.
- **B. Non-deterministic probabilistic sampling:** Correct. LLMs generate responses based on token probability distributions rather than fixed lookup tables.
- **C. Intentional anti-plagiarism randomization:** Incorrect. Duplication detection filters public code matches, rather than randomly altering variable syntax.
- **D. Admin A/B testing on seats:** Incorrect. Seat assignments do not split users into differing syntax variants.

💡 **GH-300 Exam Tip**

Because Copilot outputs are **non-deterministic**, teams must rely on automated unit tests, linters, and peer code reviews to ensure consistent code quality across developers.